

Contact

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(LinkedIn)

Top Skills

Retrieval-Augmented Generation (RAG)
Software Development
SAS (Programming Language)

Certifications

AWS Certified Machine Learning Engineer – Associate
AWS Certified Generative AI Developer - Professional Early Adopter
AWS Certified AI Practitioner
LFEL1004: Authentication & Authorization for Web/API
AWS Certified Solutions Architect – Associate

Tumelo Tshana Konaite

Data Scientist | AI & Software Engineer
Johannesburg Metropolitan Area

Summary

I help organizations transform data and AI into scalable, production-ready systems that drive measurable business impact across finance, telecom, healthcare, and digital platforms. **AI & Data Strategy:** I design and implement AI-driven solutions that move beyond experimentation into real-world operational systems — enabling smarter decision-making, automation, and business growth. **Scalable ML & GenAI Systems:** My expertise spans machine learning, NLP, generative AI, semantic retrieval (RAG), predictive analytics, and cloud-native AI architectures. I build end-to-end solutions from data ingestion and feature engineering to APIs, orchestration, and deployment. **Production-Grade Engineering:** I focus on reliability, maintainability, and scalability — applying strong software engineering principles, testing practices, and MLOps thinking to ensure AI systems perform effectively in production environments. **Risk & Quantitative Analytics:** With an MSc in Computational & Applied Mathematics, my background includes quantitative modeling, derivatives research, and risk-oriented data workflows, allowing me to bridge advanced mathematics with applied AI and analytics. **Collaboration & Technical Leadership:** I enjoy working across engineering, product, and business teams to translate complex problems into well-engineered, data-driven solutions that create tangible value. I'm passionate about building AI systems that are not only technically advanced, but practical, scalable, and impactful in the real world.

Experience

Zeno

Data Scientist | AI Engineer | Software Engineer
December 2025 - Present (6 months)
City of Johannesburg

- Architected and developed the backend for an AI-powered educational content platform, delivering production-ready APIs for automated summaries,

quizzes, flashcards, learning scorecards, and conversational learning assistants.

- Built modular generative AI pipelines using Retrieval-Augmented Generation (RAG), embeddings, transcript processing, and prompt orchestration to improve contextual relevance and response quality.
- Evaluated and benchmarked multiple foundation models and LLM providers, analysing latency, cost efficiency, and response quality to optimise model selection across different use cases.
- Designed and developed an AI-powered support agent platform that enables businesses to automate customer support workflows through Retrieval-Augmented Generation (RAG), vector search, knowledge retrieval, conversation management, and intelligent response generation.
- Implemented tenant-aware retrieval pipelines, confidence scoring, escalation workflows, observability, and evaluation mechanisms to improve reliability and production readiness.
- Architected and developed the backend for BeautyVerse, a beauty services marketplace connecting customers and service providers through booking, discovery, review, moderation, and profile management workflows.
- Built scalable FastAPI services using PostgreSQL, SQLAlchemy, Alembic, Docker, and automated testing to support maintainable and production-ready deployments.
- Implemented integrations for media storage, email notifications, calendar invite generation, and external service communication while maintaining high code quality through modular architecture and automated testing practices.

Huawei

Data Analyst | Huawei

December 2019 - December 2025 (6 years 1 month)

- Developed Python-based data pipelines and automation workflows to preprocess, transform, and analyze structured datasets, improving model readiness and operational efficiency.
- Performed exploratory data analysis (EDA), feature engineering, and statistical modeling on large datasets to uncover trends, generate insights, and support predictive decision-making.
- Designed and optimized advanced SQL queries for large-scale data extraction, transformation, and integration into machine learning pipelines and analytical dashboards.
- Led data-driven initiatives leveraging predictive analytics and performance metrics to improve project forecasting, optimize resource allocation, and increase delivery success rates.

University of the Witwatersrand
Mathematics Tutor
January 2019 - December 2019 (1 year)
South Africa

Education

University of the Witwatersrand
MSc in Computational & Applied Mathematics (2021 – 2024), Computational
and Applied Mathematics · (January 2021 - December 2024)

University of the Witwatersrand
Bachelor of Science (Hons) Mathematics, Mathematics · (2019 - 2019)

University of the Witwatersrand
Bsc, Computational and Applied Mathematics · (2016 - 2018)